

Unequal Burdens After the 2023 Maui Wildfires: Income and Access to Medical Care

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Introduction

The August 8, 2023, Maui wildfires disrupted housing, employment, transportation, and access to essential services. In the aftermath, households continued to pay for food, medications, transportation, and health care while coping with displacement and lost income. A Hawai‘i State Department of Health needs assessment found that 26% of surveyed households identified financial recovery as their greatest need and that 41% reported worsening health for at least one household member (Hawai‘i State Department of Health [HDOH], 2023). These conditions raise a distributional question: Did households with fewer financial resources have greater difficulty meeting essential needs after the fires?

Nurses are central to disaster preparedness, response, and recovery because they assess health needs, identify unmet needs, provide health education, coordinate care, and connect individuals and families with services. In recovery settings, this role extends beyond immediate clinical treatment to medication continuity, chronic disease management, transportation and referral support, and surveillance of emerging community needs. Nurses also serve as advocates who can communicate frontline evidence to public health leaders and policymakers, helping to shape policies that reduce inequities, strengthen preparedness, and support long-term community recovery (Awotona, 2017).

This paper examines whether participants in the Maui Wildfire Exposure Study (MauiWES) who lived below the federal poverty line (FPL) reported greater difficulty accessing medical care or medications than participants at or above the FPL. It also documents food insecurity among the affected cohort while distinguishing this descriptive finding from an income-stratified test that the available report cannot support. The central hypothesis is that

lower-income households faced more barriers because they had less capacity to absorb income losses, displacement, transportation costs, and other disaster-related expenses.

Two economic concepts frame the analysis. Demand for necessities, such as food, medication, transportation, and basic health care, is relatively inelastic: Households cannot simply stop needing these goods and services when income falls. Budget share is especially important because essential expenses consume a larger share of income for households with fewer resources. Consequently, the same disruption can produce more difficult trade-offs among food, prescription refills, transportation, and clinical visits for lower-income households.

The Maui wildfires also occurred in a high-cost environment. Hawai‘i’s island geography and reliance on transportation and supply systems can heighten exposure to climate-related disruptions affecting food security, transportation, and health care access (Hudson, 2016). The following analysis treats consumer prices as context and uses MauiWES access data to evaluate the income-related health care hypothesis. The hypothesis is that households below the poverty line experienced greater difficulty accessing medical care and medications during the recovery period following the Maui wildfires than households at or above the poverty line. This hypothesis would not be supported if households below the poverty line reported the same or fewer barriers to medical care than households at or above the poverty line. A separate and narrower claim—that the wildfires produced the disparity—would not be supported if the difference between income groups appeared only in barriers that existed before the fires and not in barriers that arose after them. Distinguishing these two claims matters for policy because a disparity that predates a disaster can be anticipated and screened for, whereas one created by a disaster can only be addressed afterward.

The Bureau of Labor Statistics Consumer Price Index for Food at Home in Urban Hawai‘i indicates that Maui households entered the disaster after a prolonged period of grocery price growth. Indexed to 2019, Hawai‘i food-at-home prices rose 32.9% through 2025, compared with 29.6% for the U.S. city average. More informative than the comparison with the mainland is the comparison with Hawai‘i’s own historical trend. A log-linear fit to the 2010–2019 period yields a prepandemic trend of 1.82% per year. Measured against that trend, grocery prices stood 13.7% above where the prepandemic path would have placed them at the time of the fires, widening to 17.1% by 2025 (see Figure 1). Year-over-year inflation rates understate this pressure because, by mid-2023, the sharpest increases had already passed into the comparison base. The price level, rather than the rate of change, describes what households were actually paying. This measure should not be interpreted as evidence that the wildfire raised food prices. The Urban Hawai‘i index is Honolulu-weighted and reflects a statewide rather than a Maui-specific market, and it contains no income breakdown. Thus, it documents the cost environment households faced without identifying who bore the cost. Its role here is to establish that financially constrained households entered the disaster with little slack in their budgets for the displacement, lost income, and transportation costs that followed.

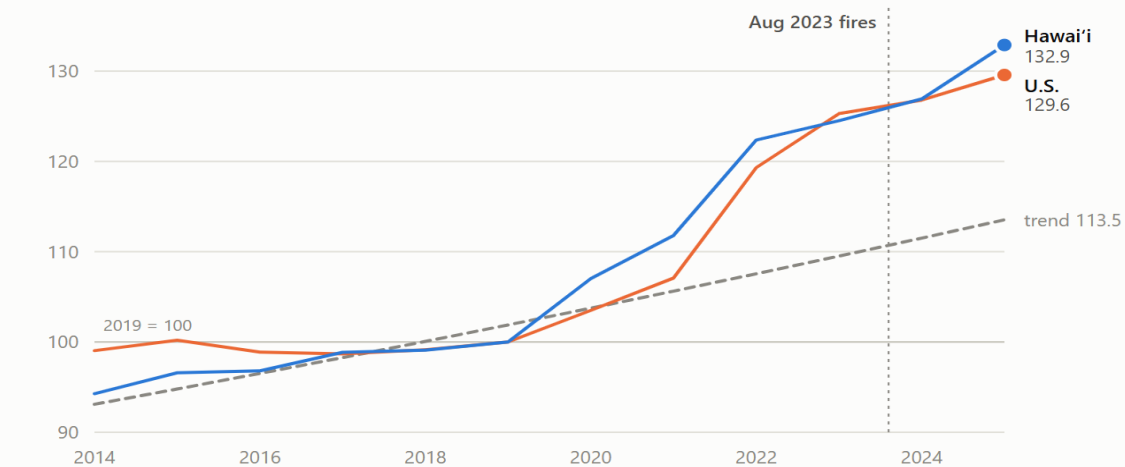
Figure 1

Food-at-Home Consumer Price Index, Urban Hawai‘i and U.S. City Average, Indexed to 2019 = 100

PANEL A - COST PRESSURE**Grocery prices in Hawai'i outran both the mainland and their own trend**

Consumer Price Index, food at home, annual averages, indexed to 2019 = 100. The dashed line extends Hawai'i's 2010–2019 log-linear trend (1.82%/yr) as a counterfactual.

— Urban Hawai'i — U.S. city average — Hawai'i 2010–2019 trend



Source: U.S. Bureau of Labor Statistics via FRED — CUUSA426SAF11 (Urban Hawai'i) and CUUR0000SAF11 (U.S. city average), 1982–84 = 100. Hawai'i CPI is published for the Urban Hawai'i CBSA, which is Honolulu-weighted; it is the closest published price series to Maui, not a Maui-specific index.

Note. Annual averages. The dashed line extends Hawai'i's 2010–2019 log-linear trend (1.82% per year) as a counterfactual. Data are from U.S. Bureau of Labor Statistics series CUUSA426SAF11 and CUUR0000SAF11, retrieved through FRED. The Urban Hawai'i CBSA series is Honolulu-weighted and is the closest published price series to Maui; it is not a Maui-specific index.

Before examining differences within the affected population, it is important to establish who comprised that population. Twenty-seven percent of MauiWES participants lived in households below the federal poverty line (95% CI [23.6%, 30.3%]). The comparable figure for Maui County as a whole is 9.2%, which is below both the Hawai'i rate of 10.0% and the national rate of 10.6% (U.S. Census Bureau, as cited in USAFacts, 2025). The fire-affected cohort was

therefore approximately 2.9 times as likely to be below the poverty line as the county overall, a difference far outside sampling variation, $z = 16.0$, $p < .001$ (see Figure 2).

Maui County is, by national standards, a low-poverty county. This makes the contrast sharper rather than weaker: The wildfires did not affect a representative cross-section of Maui. Three explanations may contribute to this gap and cannot be separated with published data. Lahaina may have housed a poorer, more working-class population than the county average. Income lost to the disaster may have pushed households below the poverty line, because MauiWES data were collected in early 2024, after the fires, and 74% of participants reported reduced household income. Finally, MauiWES is a volunteer cohort rather than a probability sample, so households in greater distress may have enrolled at higher rates. The available evidence supports the conclusion that the fire-affected cohort was markedly poorer than Maui County overall, reflecting some combination of who lived in the burn zone, income lost to the disaster, and self-selection into the study.

Figure 2

Population Below the Federal Poverty Line: Benchmarks and Fire-Affected Cohort

Note. Census benchmarks describe area populations, whereas the cohort bar describes surveyed individuals; the comparison is therefore ecological. The Maui County figure is an American Community Survey 5-year estimate spanning 2020–2024, which averages across the August 2023 fire and cannot, by design, display a fire effect. Error bars show the 95% confidence interval for the cohort estimate. Data are from U.S. Census Bureau (as cited in USAFacts, 2025) and University of Hawai‘i Economic Research Organization (UHERO, 2024).

Healthcare Access and Income

Health care access provides the clearest test of whether lower-income households experienced a disproportionate burden because MauiWES reports it separately by poverty status. Among all 679 participants, 13% reported difficulty accessing medical care or medications both before and since the fires, and an additional 28% reported difficulty that began only after the fires (UHERO, 2024). By income group, 48.1% of participants below the poverty line reported some difficulty, compared with 37.1% of those at or above it—a difference of 11.0 percentage points, 95% CI [2.6, 19.4], risk ratio (RR) = 1.30, 95% CI [1.07, 1.57], $z = 2.59$, $p = .010$. The primary hypothesis is therefore supported: Households below the poverty line reported greater difficulty obtaining care during the recovery period. Separating the two forms of difficulty, however, changes the interpretation. Among households below the poverty line, 18.0% reported barriers that existed both before and since the fires, compared with 10.1% of those at or above the poverty line, $RR = 1.79$, 95% CI [1.19, 2.68], $p = .005$. For barriers that arose only after the fires, the two groups were nearly indistinguishable: 30.1% below the poverty line and 27.0% at or above the poverty line, $RR = 1.11$, 95% CI [0.85, 1.45], $p = .43$. The income disparity in health care access is concentrated almost entirely in barriers that predated August 2023. The fires increased the absolute level of difficulty for everyone—roughly three in 10 participants in both income groups reported new barriers—but did not widen the gap between income groups. This null result should be interpreted with care. With 183 participants below the poverty line and 496 at or above it, the smallest difference this comparison could reliably detect at 80% power is a risk ratio of approximately 1.42, and the observed value was 1.11. The appropriate conclusion is that no difference was detected in newly arising barriers, not that no difference exists. These findings are consistent with the budget-share explanation: Households with fewer resources had less

capacity to absorb economic disruption while maintaining access to essential services. From a health care perspective, this demonstrates why the consequences of a disaster cannot be evaluated only by property damage or consumer prices. Economic vulnerability also becomes a health concern when limited resources coincide with difficulty obtaining basic needs, such as food.

Food Security

Food insecurity was substantial in the MauiWES cohort. Forty-seven percent of participants experienced low or very low food security, compared with 23% in the prefire Maui comparison cohort (RR = 2.01, $p < .0001$). This difference documents a serious food-security burden among affected participants and is consistent with broader evidence of household income disruption: 74% of participants reported a reduction in household income following the fires (UHERO, 2024).

However, the available report does not permit a test of whether food insecurity differed by poverty status. UHERO's published PDF cross-tabulates food security by race rather than by FPL status; the poverty-line breakdown for this item is available only through the MauiWES interactive dashboard or the underlying microdata. Therefore, food insecurity is presented here as an overall cohort outcome, not as evidence that the income hypothesis was confirmed for food security. Any figure presenting this result should identify race as the available stratifier and state this limitation in its method note.

Interpretation, Limitations, and Recommendations

Taken together, the evidence supports the income hypothesis for health care access, with an important qualification. Lower-income participants had a higher overall prevalence of difficulty accessing care or medications, and the strongest disparity was in barriers that preceded

the fires. The findings therefore support an interpretation of disaster recovery as occurring on top of unequal baseline access rather than establishing that the fires alone produced a new income-based difference in health care access.

Several limitations guide this interpretation. First, the CPI in Figure 1 is Honolulu-weighted and statewide, not Maui-specific; it provides cost context rather than a direct estimate of price changes faced by Maui households. Second, the poverty-status counts used in Table 1 are reconstructed from published percentages, although the weighted check reproduces the published overall result. Third, the observational data show associations rather than a single causal pathway. Displacement, employment changes, transportation problems, smoke exposure, psychological stress, and other factors may have contributed simultaneously to access difficulties. Finally, the available MauiWES report does not support an income-stratified analysis of food insecurity.

Table 1

Difficulty Accessing Medical Care or Medications by Poverty Status

Measure	Below FPL	Above FPL	Effect	p
Any difficulty accessing care/medications	48.1%	37.1%	RR = 1.30	.0095
Pre- existing barriers	18.0%	10.1%	RR = 1.79	.0050
New post-fire barriers	30.1%	27.0%	RR = 1.11	.4331

Note. Full distribution test: $\chi^2(2) = 10.05$, $p = .0066$. Poverty-status counts were reconstructed from published percentages. The reconstruction is internally consistent: $0.27 \times 48\% + 0.73 \times 37\% = 40.0\%$, matching the published overall rate.

These findings have a direct implication for postdisaster public health nursing. HDOH should incorporate economic vulnerability into nursing assessments and outreach, including income, displacement, transportation barriers, chronic conditions, and medication needs. Nurses can use this information to prioritize medication continuity, care navigation, transportation support, mobile services, pharmacies, and community-based assistance. Targeting preexisting access barriers is particularly important because the results indicate that recovery burdens are shaped by inequities present before a disaster.

Conclusion

The Maui wildfires demonstrate that disaster burden is shaped not only by the event itself but also by a household's ability to absorb financial shocks while meeting essential needs. Hawai'i's elevated food-cost environment increased pressure on household budgets, while MauiWES documented widespread income loss, high food insecurity, and substantial difficulty accessing medical care or medications. The evidence supports an income-related disparity in health care access, particularly for barriers that predated the fires. Because these vulnerabilities predated August 8, 2023, they can be identified before the next disaster rather than only afterward. This makes economic screening a preparedness measure, not merely a recovery measure.

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